The Augmented Synthetic Historical Control Method

Introduction

The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics for estimating treatment effects, second only to difference-in-differences. The idea behind SCM is simple: the untreated potential outcomes of a treated unit can be approximated by a weighted average of donor pool units (i.e., untreated or unexposed entities). Numerous extensions have been proposed, including singular value decomposition [Amjad et al., 2018, Costa et al., 2023], penalized regression models [Abadie and L'Hour, 2021], Bayesian methods [Fernández-Morales et al., 2025, Kim et al., 2020], and further refinements [Abadie, 2021].

Like difference-in-differences, SCM requires a suitable donor pool. In many cases, however, no appropriate donors exist—such as during global events (e.g., pandemics, financial crises) or when comparable untreated units cannot be observed. To address this, Chen et al. [2024] introduced the synthetic historical control method (SHC). Rather than relying on external donors, SHC uses the treated unit's own pre-treatment history: the pre-treatment period is split into overlapping segments, which serve as a donor pool. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, leveraging the idea that future outcomes resemble past patterns.

SHC, like Abadie et al. [2010, 2015], restricts weights to the simplex, ensuring convex combinations of donor segments. This guarantees interpretability but can be too restrictive in time-series settings where noise or complex dynamics make exact pre-treatment fit unattainable. In such cases, purely convex weights may not provide accurate approximations.

This paper introduces the *Augmented Synthetic Historical Control method (ASHC)*, which extends SHC by Chen et al. [2024] in two key ways.
First, it relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This allows for negative weights, permitting controlled extrapolation beyond the convex hull of observed segments. Second, it adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM of Ben-Michael et al. [2021]. This balances flexibility with stability: extrapolation is possible, but penalized to prevent overfitting.

ASHC also retains the stabilizing Gram penalty $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2$ from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components—the fit term, the SHC-proximity penalty, and the Gram penalty—yield a new estimator that generalizes SHC while preserving its semiparametric time-series foundation.

From a theoretical standpoint, ASHC departs from the linear factor model framework that motivates SCM. Instead, it operates entirely within a time-series context: the donor pool consists of the treated unit's own history. Our assumptions therefore follow the semiparametric regression tradition: the latent trend is smooth enough to be approximated by a truncated Taylor expansion, noise is bounded (or sub-Gaussian), and the donor matrix has a bounded operator norm. These weaker assumptions suffice to establish finite-sample error bounds, asymptotic consistency, and convergence rates for ASHC.

A natural question, echoing Ben-Michael et al. [2021], is when ASHC should be preferred to SHC. Our recommendation is empirical: the estimated SHC bias (the pre-treatment fit error) provides a direct diagnostic. If the bias is large, trading some extrapolation for better fit via ASHC is beneficial. Because this bias is measured in the same units as the treatment effect, researchers can calibrate "large" bias to the context of their application.

This paper makes two main contributions.
First, we introduce the ASHC estimator, which extends SHC by permitting limited extrapolation while penalizing instability. The estimator combines a fit term, a proximity penalty, and a Gram penalty, balancing flexibility with robustness.
Second, we establish its theoretical properties. We prove a finite-sample error bound, decomposing the error into four interpretable parts: SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. We then show that ASHC is asymptotically consistent: as pre-treatment length and smoothing order grow, prediction error converges to the noise floor, and under sub-Gaussian noise, to zero in probability. Finally, we derive the convergence rate: under optimal bandwidth, the error decays at $O_p(m^{-H/(2H+3)})$, approaching the near-parametric rate $O_p(m^{-1/2})$ as $H \to \infty$.

Together, these results demonstrate that ASHC preserves the interpretability of SHC while offering provably stronger error control and asymptotic guarantees.

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